

Abenezer Golda

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Education

The George Washington University

MASTERS OF SCIENCE IN COMPUTER SCIENCE

Washington D.C., USA

May 2027(Expected)

KIIT University

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGINEERING

Bhubaneswar, India

May 2025

Relevant Experience

Undergraduate Researcher

Bhubaneswar, India

KIIT University Mentored by Dr. Rajat Kumar and Dr. Vikas Hassija

Dec '22 - May 2025

- Proposed novel mechanisms to balance privacy guarantees and computational efficiency for big data environments.
- Analyzed modern serverless architectures for scalable, cost-effective, and real-time anomaly detection in large-scale systems.
- Executed NLP projects aimed at improving accessibility for low-resource languages, including optimizing BERT-based models and exploring novel architectures for multi-modal machine translation.
- Mentored 11 juniors, providing structured guidance on research methodologies, technical implementations, and academic writing, fostering their growth as independent researchers.

Undergraduate Research Intern

Pilani, India

BITS PILANI Mentored by Prof. Vinay Chamola

May '23 - May '24

- Conducted research on privacy-preserving deep learning techniques, focusing on safeguarding sensitive data in generative AI models.
- Proposed a multi-layered privacy framework combining secure aggregation and model regularization to mitigate security risks in generative models.
- Designed and implemented comprehensive data cleaning and preprocessing pipelines for multimodal datasets.
- Evaluated and validated methods on diverse benchmark datasets, ensuring broad applicability and generalizability.

Data Analyst

Bhubaneswar, India

KIIT IOT Lab

Mar '22 - Aug '24

- Spearheaded data analysis efforts for 10+ IoT-focused research projects, driving insights that significantly contributed to the lab's academic and experimental outcomes.
- Enhanced data analysis accuracy by 15% by developing efficient cleaning pipelines and visualizing datasets for weekly Datathon challenges.
- Designed and deployed interactive dashboards using tools like Tableau and Power BI, enabling real-time data monitoring and reporting, resulting in a 25% improvement in the accuracy and clarity of weekly analysis reports.
- Facilitated data-driven decision-making by integrating automated workflows for data collection and visualization, reducing manual reporting efforts by 20%.

Projects

Code Correctness Benchmarking & Failure-Mode Analysis for LLMs

Built a stratified 90-problem Python benchmark and automated evaluation harness to diagnose how LLMs fail at code generation, not just how often. Evaluated GPT-3.5-Turbo and CodeLlama-7B-Instruct across 540 sandboxed test runs, producing a four-category failure taxonomy (syntactic, runtime, algorithmic, edge-case) with bootstrap-validated pass@1/pass@3 metrics, revealing that 90-100% of syntactic and runtime errors are recoverable via multi-sample aggregation while algorithmic failures persist. 🔄

Multi-Lingual Video Summarizer using LLAMA 3

Developed a multi-lingual YouTube video summarizer utilizing Llama3, achieving 91.2% BLEU and 88.5% ROUGE-L scores for accuracy and context retention. With a coherence rating of 4.8/5 and a translation speed of 3.5 seconds per 1,000 tokens, it delivers scalable, responsive, and language-specific summaries for diverse users. 🔄

Low-Resource Language Chatbot

Developed a conversational AI chatbot for underrepresented languages, leveraging fine-tuned mBERT and transfer learning. Achieved 85.7% intent recognition accuracy and 88.2% BLEU score in Amharic and Tigrinya. Enabled cultural and linguistic inclusivity with a response latency of 1.9 seconds, facilitating improved accessibility for marginalized communities. 🔄

Technical Skills

Languages

Python, C, C++, Java, SQL

Software & Tools

PyTorch, TensorFlow, Django, AWS, LaTeX, Git, VS Code, Vim, PyCharm, Overleaf